GLMP promotes EGFR-TKI resistance by activating autophagy and RhoA pathway in non-small cell lung cancer

Experimental dataset evidence card

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Dataset Identity

Dataset ID: GSE298111
Title: GLMP promotes EGFR-TKI resistance by activating autophagy and RhoA pathway in non-small cell lung cancer
Taxon: Homo sapiens
Sample count: 6
Selected reason: PC9OR control versus si-GLMP replicates

Experimental Context

Assay context: resistance model; EGFR inhibition; osimertinib response
Sample accessions: GSM9008618; GSM9008621; GSM9008619; GSM9008622; GSM9008620; GSM9008623
Summary: Here, we report that the EGFR-TKI resistance mechanism is mediated by lysosome-related regulation. We established an osimertinib-resistant cell line, designated PC9OR, derived from the parental PC9 cells. The overexpression of glycosylated lysosomal membrane protein (GLMP) in PC9OR promotes resistance to Osimertinib. Mechanistically, GLMP could regulate the ubiquitination of RhoA and promote resistance by activating the epithelial-mesenchymal transition (EMT). Our findings provide a potential therapeutic strategy to overcome resistance to EGFR-TKIs.

Provenance

Provenance: derived_from_raw_layer
Privacy posture: cell-line or assay-level source selected
Raw source trace: 00_raw/json/datasets/geo/GSE298111/geo_esummary.json; 00_raw/txt/datasets/geo/GSE298111/series_soft.txt; 00_raw/json/datasets/geo/GSE298111/source_record.json